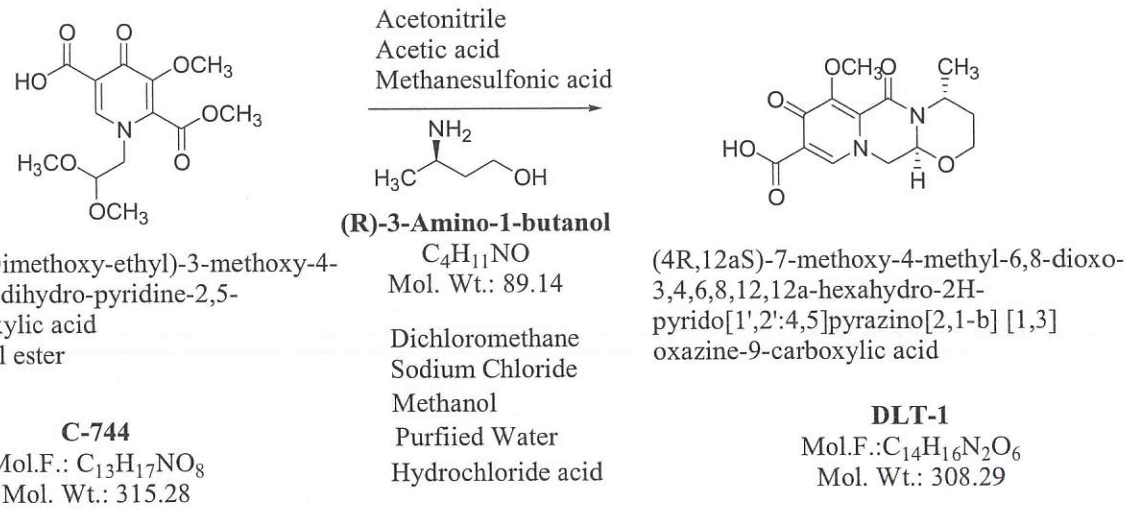
DOLUTEGRAVIR SODIUM ROS

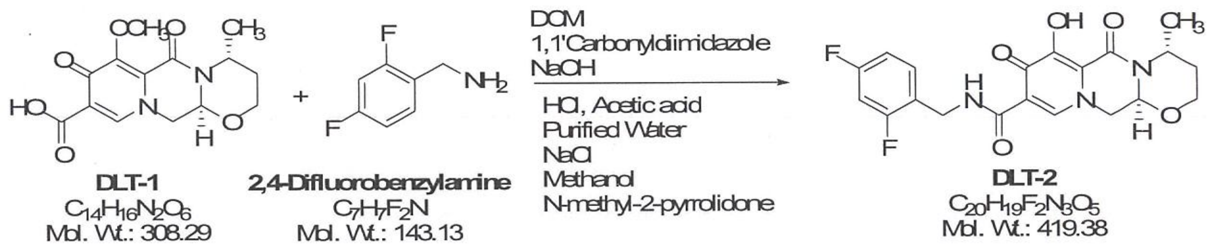
ANNEXURE-1

Dolutegravir Sodium involves three stages. ROS and Brief Process details are detailed below

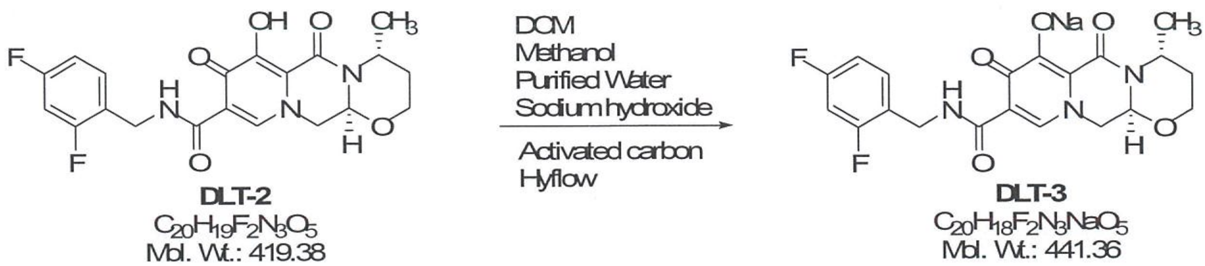
DLT-1: Preparation of (4R,12aS)-7-methoxy-4-methyl-6,8-dioxo-3,4,6,8,12,12a-hexahydro-2H-pyrido[1',2':4,5]pyrazino[2,1-b][1,3]oxazine-9-carboxylic acid (DLT-1)



DLT-2: Preparation of (4R,12aS)-N-[(2,4-Difluorophenyl)methyl]-3,4,6,8,12,12a-hexahydro-7-hydroxy-4-methyl-6,8-dioxo-2H-pyrido[1',2':4,5]pyrazino[2,1-b][1,3]oxazine-9-carboxamide (Dolutegravir / DLT-2)



DLT-3:Preparation of Sodium (4*R*,12a*S*)-9-{[(2,4-Difluorophenyl)methyl]carbamoyl}-4-methyl-6,8-dioxo-3,4,6,8,12,12*a*-hexahydro-2*H*-pyrido[1′,2′:4,5]pyrazino[2,1-*b*] [1,3]oxazol-7-olate (Dolutegravir sodium / DLT-3)



DOLUTEGRAVIR SODIUM ROS

ANNEXURE-1

DLT-1: Preparation of (4R,12aS)-7-methoxy-4-methyl-6,8-dioxo-3,4,6,8,12,12a-hexahydro-2*H*-pyrido[1',2':4,5]pyrazino[2,1-b][1,3]oxazine-9-carboxylic acid (DLT-1).

1-(2,2-Dimethoxyethyl)-5-methoxy-6-(methoxycarbonyl)-4-oxo-1,4-dihydropyridine-carboxylic acid (Stage-2) in acetonitrile on reaction with acetic acid and methanesulfonic acid to yield pyridinone carboxylic acid aldehyde (3A) and is in-situ reaction with (R)-amino-1-butanol to afford (4R,12aS)-7-methoxy-4-methyl-6,8-dioxo-3,4,6,8,12,12a-hexahydro-2*H*-pyrido[1',2':4,5]pyrazino[2,1-b] [1,3] oxazine-9-carboxylic acid (**DLT-1**).

DLT-2: Preparation of (4R,12aS)-N-(2,4-difluorobenzyl)-7-methoxy-4-methyl-6,8-dioxo-3,4,6,8,12,12a-hexahydro-2*H*-pyrido[1',2':4,5]pyrazino[2,1-b][1,3]oxazine-9-carboxamide (DLT-2).

(4R,12aS)-7-methoxy-4-methyl-6,8-dioxo-3,4,6,8,12,12a-hexahydro-2*H*-yrido[1',2':4,5]pyrazino[2,1-b][1,3] oxazine-9-carboxylic acid (DLT-1) in Dichloromethane on reaction with 2,4-Difluorobenzylamine in the presence of a coupling reagent 1,1'-carbonyldiimidazole (CDI) to afford (4R,12aS)-N-(2,4-difluorobenzyl)-7-methoxy-4-methyl-6,8-dioxo-3,4,6,8,12,12a-hexahydro-2*H*-pyrido[1',2':4,5] pyrazino[2,1-b][1,3]oxazine-9-carboxamide (Methoxy Dolutegravir). Methoxy Dolutegravir reaction with Sodium Hydroxide convert to Dolutegravir sodium. Sodium cleavage with Acetic acid to afford (4R,12aS)-N-[(2,4-Difluorophenyl)methyl]-3,4,6,8,12,12a-hexahydro-7-hydroxy-4-methyl-6,8-dioxo-2*H*-pyrido [1',2':4,5]pyrazino[2,1-b][1,3]oxazine-9-carboxamide (Dolutegravir) and is isolated in a mixture of N-methyl-2-prrolidone and methanol.

DLT-3: Preparation of Sodium (4R,12aS)-9-{[(2,4-difluorophenyl) methyl]carbamoyl}-4-methyl-6,8-dioxo-3,4,6,8,12,12a-hexahydro-2H-pyrido[1′,2′:4,5] pyrazino[2,1-b][1,3]oxazol-7-olate (Dolutegravir sodium / DLT-3).

4R,12aS)-N-[(2,4-Difluorophenyl)methyl]-3,4,6,8,12,12a-hexahydro-7-hydroxy-4-methyl-6,8-dioxo-2*H*-pyrido [1',2':4,5]pyrazino[2,1-b][1,3]oxazine-9-carboxamide (Dolutegravir) is dissolved in a mixture of methanol and DCM and is treated with activated carbon to get particle free. The clear filtrate is concentrated under vacuum to get solid. To the obtained solid added methanol and is heated to reflux, to the resulting slurry added aqueous sodium hydroxide solution at reflux and is cooled to 20-25°C to get Dolutegravir sodium (DLT-3).

Authorized By

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